

AUDITABLE BRANCH CLOSURE FOR HUMAN-AI PDE PROOFS

GUILLEM DURAN BALLESTER

ABSTRACT. We introduce branch closure, a human-AI methodology for developing long PDE proofs as finite, auditable case decompositions. A branch records its local space-time window, analytic hypotheses, gauge and compactness choices, active case, goal, and supporting evidence. Every transition must be both locally justified and exhaustive. This yields a simple soundness theorem and, before all branches close, a precise partial result whose only possible counterexamples lie in the open branches.

The central repair operation differs from dependency-based invalidation. If a step asserting a predicate P fails, the last verified branch is partitioned into P and $\neg P$. All subsequent arguments are retained verbatim on the P -branch; only the complementary branch is new. For a conjunctive claim, ordered first-failure branches identify the first missing hypothesis. This makes proof repair monotone and gives an AI system a bounded task without weakening mathematical completeness.

We give a complete finite-energy Stokes example in which a compact-source pressure split leaves a spatially harmonic tail. Solenoidal cutoff estimates regularize the velocity, a local Caccioppoli contradiction excludes the parasitic harmonic branch, and the equation closes the pressure tail without a whole-space pressure or heat-kernel formula. The protocol is then instantiated in three companion developments for the three-dimensional Navier-Stokes equations: a local concentration and Seregin-profile reduction for the Type I regime, a Liouville closure of the residual Type I profile class, and a local exclusion of the complementary Type II regime. Together, these developments exhaust the local Type I/Type II dichotomy and rule out finite-time singular points for finite-energy suitable weak solutions on $\mathbb{R}^3$.

1. INTRODUCTION

Long proofs in nonlinear PDE are difficult to develop with language models for a reason that is only partly linguistic. Their conclusions depend on choices that remain implicit in ordinary prose: the cylinder on which an estimate is valid, the scale and frame, the pressure gauge, the quotient representative, the topology used at a compactness limit, and the alternative left after an estimate fails. A locally plausible paragraph can silently change one of these choices and thereby change the theorem.

Current AI systems for mathematical reasoning emphasize formal proof search, autoformalization, retrieval, and repair of rejected proof scripts [WJL⁺22, JWZ⁺22, FRRB23, LDZ24]. Recent PDE-facing systems instead translate control requests or generate numerical solvers [SSX⁺25, LMS⁺25]. These are important but different problems. The present paper addresses the intermediate activity in which a mathematician and an AI system jointly construct a rigorous analytic argument whose decisive estimates may remain in conventional mathematical language.

Our proposal is to represent the proof as an exhaustive directed acyclic graph of analytic cases. Nodes are not fragments of prose. Each node exposes the state required to interpret its claim; each edge records one implication or one exhaustive split; each terminal records a proof of the target on that case. The representation is especially natural for concentration compactness, profile decompositions, local pressure splittings, scale selection, and rigidity arguments, where the analysis already advances by alternatives [Lio84a, Lio84b, Gér98, BG99, Ker01].

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The key point is how the graph changes when an intermediate claim is wrong. Suppose a node was intended to establish properties X and Y , and every later step uses only the declared interface $X \wedge Y$. Discovering that the attempted proof does not establish Y does *not* erase the later mathematics. The preceding case is replaced by the exhaustive alternatives

$$\neg X, \quad X \wedge \neg Y, \quad X \wedge Y.$$

The existing continuation is attached unchanged to the third alternative. The first two alternatives are new analytic obligations. The root theorem is complete once they close; until then, they describe exactly what remains. We call this operation *branch-preserving repair*.

The paper makes four contributions.

- (1) It defines a PDE-specific branch state and an interface discipline for human–AI proof development.
- (2) It proves soundness, an exact partial-closure statement, and branch-preserving and monotone repair results.
- (3) It gives an operational division of labor: AI proposes and maintains local branches, while the mathematician controls theorem statements, admissible imports, and the acceptance of analytic evidence.
- (4) It supplies a complete finite-energy Stokes instantiation that closes a harmonic pressure tail by local estimates and contradiction, and points to three full-scale Navier–Stokes proof graphs using the same reading-guide, source-ledger, monotonicity-ledger, and branch-closure discipline.

The method is deliberately lighter than formalization. It does not certify an unproved estimate, and graph consistency is not a substitute for analysis. Its purpose is to make the analytic obligations explicit, preserve valid work under repair, and expose the exact residual theorem at every stage.

2. BRANCH-COMPLETE PROOF STATES

2.1. Semantic setting. Fix a universe $\mathcal{U}$ of admissible mathematical objects. An element may be a weak solution, a rescaled sequence, a limiting profile, or a local space–time configuration. A theorem has the form

$$A(s) \implies T(s), \quad s \in \mathcal{U}, \quad (2.1)$$

where A is the theorem hypothesis and T its conclusion. Direct and contradiction arguments are both included: in the latter case, the active goal is the emptiness of $A \wedge \neg T$.

Definition 2.1 (PDE branch state). A branch state is a record

$$B = (W, H, \Gamma, \tau, O, E).$$

Here W fixes the local window, scale, and coordinate frame; H is the list of analytic hypotheses already available on that window; Γ records gauge, quotient, normalization, and convergence choices; τ is the active case predicate; O is the goal on this case; and E contains the mathematical evidence accepted for the preceding implications. Its semantic region is

$$\llbracket B \rrbracket = \{s \in \mathcal{U} : W(s) \wedge H(s) \wedge \Gamma(s) \wedge \tau(s)\}.$$

The fields are mathematical data. They may be written in prose, equations, or a table; no particular serialization is part of the method.

The separation between H , Γ , and τ matters in PDE. A bound can survive rescaling while a gauge choice does not; a sequence can converge weakly in the energy space without giving the strong convergence needed for a nonlinear term; and a pressure statement can be meaningful only modulo functions of time. Recording these choices prevents a downstream argument from consuming a stronger statement than its parent produced.

Definition 2.2 (Valid transition). A transition from a branch B to children $B_1, \dots, B_m$, written

$$B \Longrightarrow B_1 \vee \dots \vee B_m,$$

is valid if its evidence establishes

$$\llbracket B_i \rrbracket \subseteq \llbracket B \rrbracket \quad (1 \leq i \leq m), \quad \llbracket B \rrbracket \subseteq \bigcup_{i=1}^m \llbracket B_i \rrbracket. \quad (2.2)$$

The first condition is refinement and the second is exhaustiveness. Children may overlap, although disjoint alternatives are preferable for diagnosis. A one-child transition is an implication; a multi-child transition is a case split.

Every transition also declares an *output interface*: the facts that its children may use. A proof below the transition is interface-respecting when it uses the transition only through those facts, not through an unstated detail of the argument that produced them. This familiar modularity condition is what makes branch-preserving repair possible.

Definition 2.3 (Closed proof graph). A finite rooted directed acyclic graph is branch-complete for (2.1) if:

- (1) $\{s : A(s)\} \subseteq \llbracket B_0 \rrbracket$ for its root B_0 ;
- (2) every nonterminal node has a valid transition to all of its children;
- (3) every terminal L has accepted evidence for $\llbracket L \rrbracket \subseteq \{s : T(s)\}$.

A terminal may close by a direct estimate, a contradiction, a reduction to an earlier theorem, or identification with an explicitly classified exceptional case.

The graph is a proof presentation, not a second mathematical theory. Its benefit is that coverage, local hypotheses, and sources become reviewable separately from the prose surrounding them.

2.2. PDE transitions. The states used in this paper admit five recurring transitions.

- (1) A *localization* replaces a global object by nested cylinders or frequency windows and records every cutoff error.
- (2) A *decomposition* splits velocity, pressure, profiles, scales, or defects into named components and retains the remainder.
- (3) An *estimate* closes the component covered by a theorem whose hypotheses appear in the current branch.
- (4) A *compactness or rigidity step* changes the state only in the topology actually obtained and records the property transported to the limit.
- (5) A *decision* introduces exhaustive positive and negative cases for a property not yet known.

For each node a reader should be able to answer six questions: what case is active, what inputs are available, what new fact is proved, which result proves it, which alternatives leave the node, and where each alternative closes. In the Stokes example these questions are answered by [Table 3](#); the larger Navier–Stokes instances use the same format.

2.3. Human–AI division of labor.

Protocol 2.4 (Branch-closure loop). Starting from (2.1), the collaborators repeat the following steps.

- (1) The mathematician fixes the theorem, admissible solution class, and which external results may be cited.
- (2) The AI proposes the next implication or exhaustive split, together with its input and output interfaces and a source-level argument.

- (3) The mathematician accepts, rejects, or weakens that argument. Simple graph checks may verify numbering, acyclicity, coverage declarations, and source links, but acceptance of an analytic estimate remains mathematical review.
- (4) An accepted transition is added. A rejected assertion P is replaced by the split $P \vee \neg P$; the existing continuation is retained on P , and work proceeds on $\neg P$.
- (5) The process stops only when every reachable branch has a terminal proof of the target.

This allocation uses the model where it is effective: proposing decompositions, searching for missing cases, restating local obligations, tracing citations, and maintaining a large graph. It leaves theorem ownership and analytic judgment with the human author. In particular, a fluent proof sketch is never itself evidence that a branch is closed.

3. CORRECTNESS AND REPAIR

Theorem 3.1 (Soundness). *If a finite graph is branch-complete for (2.1), then $A(s) \Rightarrow T(s)$ for every $s \in \mathcal{U}$.*

Proof. Fix s with $A(s)$. The root condition gives $s \in \llbracket B_0 \rrbracket$. At every nonterminal node containing s , exhaustiveness in (2.2) gives a child containing s . Finiteness and acyclicity produce a terminal L with $s \in \llbracket L \rrbracket$. Terminal closure then gives $T(s)$. $\square$

The same argument describes an incomplete proof exactly.

Theorem 3.2 (Partial closure). *Suppose every transition in a finite rooted graph is valid, but a set $\mathcal{L}_{\text{open}}$ of reachable leaves has not yet been closed. If all other leaves prove T , then*

$$\{s : A(s) \wedge \neg T(s)\} \subseteq \bigcup_{L \in \mathcal{L}_{\text{open}}} \llbracket L \rrbracket. \quad (3.1)$$

Thus the open leaves, rather than an informal list of gaps, are the exact residual classes in which a counterexample may remain.

Proof. Route a putative s with $A(s) \wedge \neg T(s)$ from the root as in the proof of Theorem 3.1. It cannot reach a closed leaf, so it reaches an open one. $\square$

3.1. Branch-preserving repair. Dependency-graph language often suggests that a failed lemma invalidates its forward dependency closure. That is not the repair rule here. The downstream proofs are conditional arguments, and a failed attempt to establish their input does not make those conditional arguments false.

Theorem 3.3 (Branch-preserving repair). *Let B be the last verified branch before a transition intended to establish a predicate P . Assume every descendant of that transition is interface-respecting: it uses the transition only through P and the facts already present in B . If the evidence for P fails, replace the transition by*

$$B \Longrightarrow B^+ \vee B^-, \quad \llbracket B^+ \rrbracket = \llbracket B \rrbracket \cap \{P\}, \quad \llbracket B^- \rrbracket = \llbracket B \rrbracket \cap \{\neg P\}. \quad (3.2)$$

Then:

- (1) *the replacement transition is valid independently of the failed evidence;*
- (2) *the entire former descendant graph remains valid, without alteration, below B^+ ;*
- (3) *the only new proof obligation is B^- ; and*
- (4) *if the former descendant graph closed B^+ , the repaired graph satisfies (3.1) with B^- as its additional residual class.*

Proof. The two sets in (3.2) refine $\llbracket B \rrbracket$ and cover it by excluded middle. On B^+ , the predicate P is part of the case definition. Every former descendant therefore receives exactly the interface it previously consumed, so its proof is unchanged. No claim has been made on B^- ; it is the new open leaf. The final assertion is Theorem 3.2. $\square$

Figure 1 shows the repaired local graph. The descendant subgraph is not regenerated; it is reattached to the case in which its declared input P holds.

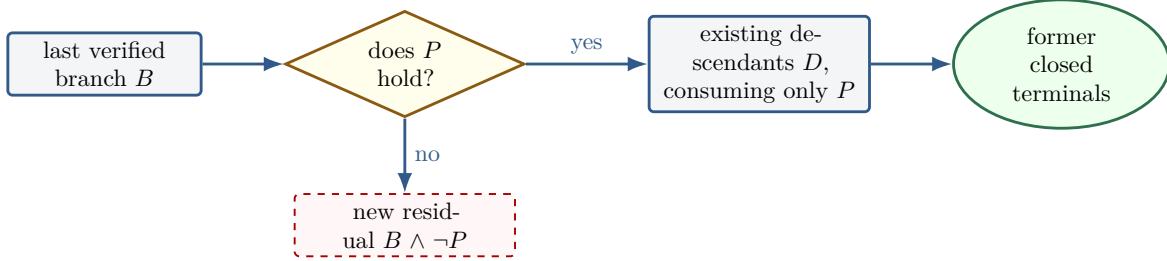


FIGURE 1. Branch-preserving repair. The positive arm retains every interface-respecting descendant; the negative arm is the only new analytic obligation.

Corollary 3.4 (Ordered first-failure repair). *If the failed node asserted $P = X_1 \wedge \dots \wedge X_m$, it may be replaced by the disjoint exhaustive split*

$$\begin{aligned}
 &\neg X_1, \\
 &X_1 \wedge \neg X_2, \\
 &\vdots \\
 &X_1 \wedge \dots \wedge X_{m-1} \wedge \neg X_m, \\
 &X_1 \wedge \dots \wedge X_m.
 \end{aligned} \tag{3.3}$$

The former descendants remain on the final branch. Each new branch records the first absent property and can be analyzed independently.

For the motivating two-property example, discovering that Y is not established creates the case $X \wedge \neg Y$; it does not alter any theorem proved under $X \wedge Y$. The additional case $\neg X$ is retained when X was also part of the unproved conjunctive node. If X was already verified earlier, that branch is empty and may be closed immediately.

Theorem 3.5 (Monotone closure). *Assume repairs are made by Theorem 3.3 and accepted evidence is changed only when its own statement or proof is found defective. Closing or refining one open branch does not reopen any other closed branch. If every newly created complementary branch is eventually closed, the resulting graph proves the original theorem.*

Proof. A repair partitions one semantic region and leaves all other regions and their evidence unchanged. Refinement preserves this separation. When all leaves are closed, Theorem 3.1 applies. $\square$

The role of dependency information is therefore diagnostic rather than destructive. It identifies the interface at which to insert the dichotomy and the continuation to attach to its positive arm. It does not mark a forward set of mathematical statements as invalid.

4. SPECIALIZATION TO PDE PROOFS

For a PDE argument, the abstract record in [Definition 2.1](#) is made concrete as follows.

TABLE 1. Minimal information carried by a PDE branch.

Field	Typical content	Question answered
W	Cylinder, cutoff, scale, frame, time orientation	Where is the claim valid?
H	Equation, solution class, bounds, sign or flux assumptions	Which analytic hypotheses are available?
Γ	Pressure gauge, quotient, normalization, convergence topology	Which representative and limiting meaning are fixed?
τ	Profile type, scale relation, active defect, positive/negative case	Which alternative is being analyzed?
O	Regularity, decay, contradiction, classification, or reduction	What must this branch prove?
E	Lemma, calculation, citation, and exact input/output interface	Why may the reader traverse the incoming edge?

Three constraints are particularly important.

Locality. An interior estimate consumes data on an explicitly larger window and produces a conclusion on an explicitly smaller one. Cutoff errors and pressure remainders are children of the same node unless a cited estimate controls them.

Representative stability. Statements involving pressure, harmonic gradients, potentials, or quotient spaces specify the representative that survives the transition. Equality of vorticity does not, by itself, identify velocity; weak convergence of pressure does not choose its time-dependent additive constant.

Topological sufficiency. Each limit passage names the convergence needed by the next nonlinear operation. A missing strong convergence statement is handled as a negative branch, not silently upgraded from a weak one.

These constraints turn common review questions into branch questions. If a profile decomposition omits a scale relation, split on that relation. If a pressure estimate controls only the nonharmonic component, retain the harmonic component as a branch. If a rigidity lemma requires $X \wedge Y$, use the ordered alternatives in (3.3). In each case the successful continuation remains available while the complementary analytic class becomes the next research target.

5. COMPLETE STOKES INSTANTIATION

The example now isolates the nonlocal pressure issue that occurs in the Navier–Stokes developments while removing the nonlinear term. On $\mathbb{R}^3 \times (0, T)$ consider

$$\partial_t u - \Delta u + \nabla p = -\operatorname{div} F, \quad \operatorname{div} u = 0, \quad (5.1)$$

where $(\operatorname{div} F)_i = \partial_j F_{ij}$. The pressure is defined modulo a distribution depending only on time.

Definition 5.1 (Finite-energy Stokes state). A finite-energy weak solution of (5.1) satisfies

$$u \in L_{\operatorname{loc}}^\infty(0, T; L^2(\mathbb{R}^3)) \cap L_{\operatorname{loc}}^2(0, T; \dot{H}^1(\mathbb{R}^3)), \quad p \in \mathcal{D}'(\mathbb{R}^3 \times (0, T)),$$

and the equation holds in distributions. We assume $F \in C^\infty(\mathbb{R}^3 \times (0, T))$, with its derivatives locally square integrable in time and space. This is the linear analogue of the finite-energy class used in the Navier–Stokes applications.

Fix $B_r \Subset B_\rho \Subset B_R$, choose $\chi \in C_c^\infty(B_R)$ with $\chi = 1$ on B_ρ , and let $N(x) = (4\pi|x|)^{-1}$, so $-\Delta N = \delta_0$. Define the compact-source pressure and the pressure tail by

$$p_{\text{near}} = \partial_i \partial_j N * (\chi F_{ij}), \quad p_{\text{tail}} = p - p_{\text{near}} - c(t), \quad (5.2)$$

where $c(t)$ fixes the spatial mean on B_ρ . Then $\Delta p_{\text{tail}} = 0$ on B_ρ . The tail contains all influence not represented by the compactly supported local source; its regularity will be proved, not assumed.

Remark 5.2 (Why finite energy is essential). For a nonsmooth scalar function $a(t)$,

$$u(x, t) = a(t)e_1, \quad p(x, t) = -a'(t)x_1, \quad F = 0$$

is a local distributional Stokes solution with zero vorticity. It is excluded from [Definition 5.1](#) unless $a = 0$, because a nonzero spatially constant velocity is not in $L^2(\mathbb{R}^3)$. Thus a local curl argument alone cannot prove the theorem below; the finite-energy branch must be used.

Theorem 5.3 (Finite-energy Stokes regularity by local pressure-tail closure). *Let (u, p, F) be a finite-energy Stokes state on $\mathbb{R}^3 \times (0, T)$. Then*

$$u \in C^\infty(\mathbb{R}^3 \times (0, T)), \quad \nabla p \in C^\infty(\mathbb{R}^3 \times (0, T)).$$

For every compact cylinder $Q' \Subset \mathbb{R}^3 \times (0, T)$ and every derivative order, the corresponding norms of u and ∇p are bounded by the finite-energy norm of u on a larger time interval and finitely many local norms of derivatives of F on a buffered spatial cylinder. For every split (5.2), $\nabla p_{\text{tail}} \in C^\infty(B_r \times I_r)$.

The proof uses no whole-space pressure formula and no global heat or Oseen kernel. Compactly supported divergence corrections remove pressure from the velocity estimates. The only possible failure is a finite-energy parasitic harmonic mode; local Caccioppoli estimates on expanding balls force that mode to vanish. Once the velocity is smooth, the equation closes the harmonic pressure tail locally.

5.1. How to read the proof architecture. Node [2] records the local near/tail pressure split without assigning any time regularity to the harmonic tail. Curl and local heat estimates give the smooth vorticity in [3]. Solenoidal cutoffs and annular Bogovskiĭ corrections then yield the exhaustive alternative [4]: either the local velocity bootstrap closes, or its first failure produces a nonzero entire finite-energy field with zero divergence and curl. The failure branch reaches [6], where an expanding-ball Caccioppoli estimate contradicts its normalization. On the surviving branch, [8] uses the original equation to prove smoothness of the pressure tail, and [9] gives full regularity of the unmodified velocity and pressure gradient.

5.2. Diagram map. [Table 2](#) records the single diagram, its node range, and the branch closed by the local contradiction.

TABLE 2. Map of the Stokes proof-flow diagram.

Diagram	Nodes	Role
Figure 2	[1]–[9]	Split the pressure locally, close the only parasitic velocity branch by finite-energy contradiction, and recover the harmonic pressure tail from the equation.

5.3. Proof-dependency diagram. [Figure 2](#) displays the local analytic spine and both outputs of the solenoidal-bootstrap decision.

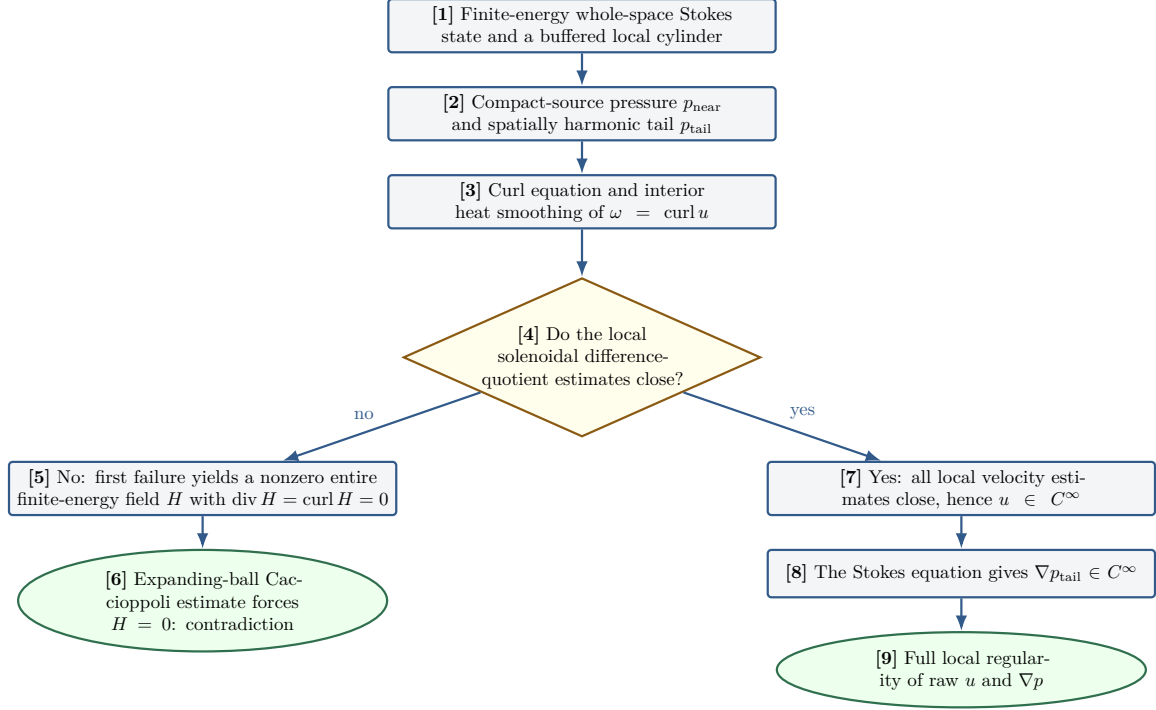


FIGURE 2. Local closure of the Stokes pressure tail. The only negative branch is converted into a finite-energy harmonic mode and closed by contradiction; the positive branch recovers the unmodified pressure from the equation.

5.4. **Node-by-node audit table.** Table 3 identifies the precise lemma supporting every state, decision, and terminal closure.

Table 3: Source ledger for Figure 2.

Node	Input	Output	Mathematical source
[1]	Definition 5.1 and a target cylinder	A finite-energy state with fixed local buffers	Hypotheses of Theorem 5.3.
[2]	[1] and a spatial cutoff	$p = p_{\text{near}} + p_{\text{tail}} + c(t)$, with $\Delta p_{\text{tail}} = 0$ locally	Lemma 5.4.
[3]	[1]	Smooth local vorticity	Lemma 5.5.
[4]	[1], [3], nested cylinders	Exhaustive success/failure alternative for the local solenoidal bootstrap	Lemma 5.7.
[5]	Failure at [4]	A normalized nonzero entire L^2 field with zero divergence and curl	Failure alternative in Lemma 5.7.
[6]	[5]	Contradiction	Lemma 5.6.
[7]	Success at [4]	$u \in C^\infty$ on the target cylinder	Iteration in Lemma 5.7.
[8]	[2], [7]	Smooth harmonic pressure tail	Lemma 5.8.
[9]	[2], [7], [8]	Smooth raw velocity and pressure gradient	Proof of Theorem 5.3.

5.5. Monotone-hypothesis ledger. Table 4 records the data retained while cylinders shrink and the failure branch is exhausted over larger balls.

TABLE 4. Hypotheses retained along the Stokes route.

Datum	How it is used	Persistence
Nested cylinders	Support cutoffs, annular divergence corrections, and interior estimates	Windows only shrink.
Finite-energy bound for u	Controls every cutoff estimate and excludes the parasitic kernel	Retained on every branch and every larger ball.
Smooth tensor forcing F	Supplies the local pressure source and the difference-quotient right-hand sides	Differentiated only to finite order.
Distributional Stokes equation and incompressibility	Remove pressure from solenoidal tests and recover it after velocity closure	Never replaced by a projected whole-space equation.
Pressure modulo functions of time	Makes the conclusion a statement about ∇p	Fixed in (5.2).

5.6. Branch-closure ledger. Table 5 verifies that the success and failure alternatives both reach a proof of the original theorem.

TABLE 5. Exhaustive closure of the local Stokes bootstrap.

Case from [4]	Terminal	Closure
The local estimates fail	[6]	The normalized first-failure mode is entire, finite-energy, divergence-free, and curl-free; expanding-ball Caccioppoli forces it to vanish, contradicting normalization.
The local estimates close	[9]	Iteration makes u smooth; the original equation then makes ∇p_{tail} , and hence ∇p , smooth on the smaller cylinder.

5.7. Analytic proof. All estimates in this subsection are obtained on nested cylinders. The whole-space hypothesis supplies only the finite-energy bound that excludes the parasitic kernel; no whole-space pressure representation or heat kernel is used.

Lemma 5.4 (Localized pressure splitting). *Let $B_r \Subset B_\rho \Subset B_R$, let $\chi \in C_c^\infty(B_R)$ equal one on B_ρ , and define p_{near} and p_{tail} by (5.2). Then*

$$-\Delta p_{\text{near}} = \partial_i \partial_j (\chi F_{ij}) \quad \text{on } \mathbb{R}^3, \quad \Delta p_{\text{tail}} = 0 \quad \text{on } B_\rho.$$

For every spatial derivative order, the local norms of p_{near} on B_ρ are bounded by finitely many local norms of F on B_R .

Proof. Taking divergence in (5.1) gives

$$-\Delta p = \partial_i \partial_j F_{ij}. \tag{5.3}$$

Since $-\Delta N = \delta_0$, the compact-source potential in (5.2) satisfies the first displayed identity. Because $\chi = 1$ on B_ρ , subtraction from (5.3) gives the second. Calderón–Zygmund and interior elliptic estimates give the local bounds [CZ52, Ste70]. The construction uses the Newtonian parametrix only on the compactly supported source χF ; it makes no assertion about a whole-space representative of p . $\square$

Lemma 5.5 (Vorticity equation and local smoothing). *For $\omega = \operatorname{curl} u$,*

$$(\partial_t - \Delta)\omega = -\operatorname{curl} \operatorname{div} F \quad (5.4)$$

in distributions. Hence ω is smooth on every compact cylinder in $\mathbb{R}^3 \times (0, T)$.

Proof. Apply curl to (5.1); the pressure term vanishes. Interior hypoellipticity for the heat operator on nested cylinders gives smoothness because F is smooth. This is a local parametrix argument: a cutoff is inserted first, and every commutator is supported away from the target cylinder [LSU68, Chapter IV]. $\square$

Lemma 5.6 (Finite-energy harmonic kernel). *If $H \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ satisfies*

$$\operatorname{div} H = 0, \quad \operatorname{curl} H = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^3),$$

then $H = 0$.

Proof. The vector identity $-\Delta H = \operatorname{curl} \operatorname{curl} H - \nabla \operatorname{div} H$ shows that every component of H is harmonic. Let η_R be one on B_R , supported in B_{2R} , and satisfy $|\nabla \eta_R| \leq C/R$. Testing the harmonic equation with $\eta_R^2 H$ gives the local Caccioppoli estimate

$$\int_{B_R} |\nabla H|^2 \leq \frac{C}{R^2} \int_{B_{2R}} |H|^2 \leq \frac{C}{R^2} \|H\|_{L^2(\mathbb{R}^3)}^2.$$

Letting $R \rightarrow \infty$ yields $\nabla H = 0$. An $L^2(\mathbb{R}^3)$ constant field is zero. $\square$

Lemma 5.7 (Local solenoidal bootstrap and its failure mode). *Let (u, p, F) satisfy Definition 5.1. For nested cylinders $Q_r = B_r \times J_r \Subset B_R \times J_R = Q_R$ and every integer $m \geq 0$, the derivatives of u of parabolic order at most m satisfy an estimate of the form*

$$\sum_{2j+|\alpha| \leq m} \|\partial_t^j \partial_x^\alpha u\|_{L^2(Q_r)} \leq C_{m,r,R,J_r,J_R} \left(\|u\|_{L^2(J_R; L^2(\mathbb{R}^3))} + \mathcal{N}_{m,R,J_R}(F) \right), \quad (5.5)$$

where $\mathcal{N}_{m,R,J_R}(F)$ contains finitely many local L^2 -norms of derivatives of F on Q_R . Consequently $u \in C^\infty(Q_r)$.

In the contradiction proof of (5.5), failure at the first derivative order produces, after normalization and exhaustion by balls, a field

$$0 \neq H \in L^2(\mathbb{R}^3; \mathbb{R}^3), \quad \operatorname{div} H = 0, \quad \operatorname{curl} H = 0.$$

Thus the failure alternative is excluded by Lemma 5.6.

Proof. Choose spatial and temporal cutoffs supported in Q_R and equal to one on an intermediate cylinder. Spatial difference quotients of the weak equation cannot be tested directly after multiplication by a cutoff, because the cutoff destroys incompressibility. On the annulus where the cutoff varies, solve

$$\operatorname{div} w = \nabla \eta \cdot D_h u, \quad w|_{\partial A} = 0,$$

with the Bogovskii operator. The corrected test field $\eta D_h u - w$ is compactly supported and divergence-free. Testing the Steklov-averaged equation with this field removes the pressure identically. The local energy estimate, the boundedness of the annular divergence correction, and Young's inequality give the first spatial difference-quotient bound. Repeating the argument for differentiated equations and using (5.4) supplies all spatial derivatives. Temporal difference quotients are then obtained from the same solenoidal weak equation; the spatial estimates absorb the Laplacian term. Iteration over strictly nested cylinders proves (5.5). This is the local cutoff proof of the standard finite-energy Stokes regularity estimate; see [Soh01, Chapters II–III].

For completeness, consider the first stage at which this iteration could fail. Normalize the failed difference quotients by their first uncontrolled local norm and repeat the cutoff estimate on B_R for successively larger R . The normalized forcing tends to zero, while Lemma 5.5 makes the normalized curls tend to zero on every fixed ball. The divergence remains zero. Weak compactness in the

finite-energy space and a diagonal extraction therefore produce an entire limit H with zero divergence and curl. The normalization makes its $L^2(B_1)$ -norm nonzero, and the inherited energy bound gives $H \in L^2(\mathbb{R}^3)$. This is the displayed failure mode. It contradicts [Lemma 5.6](#), so every stage of the iteration closes. $\square$

Lemma 5.8 (Local closure of the harmonic pressure tail). *On the cylinders of [Lemma 5.4](#),*

$$\nabla p_{\text{tail}} = -\operatorname{div} F - \partial_t u + \Delta u - \nabla p_{\text{near}}.$$

Consequently ∇p_{tail} is smooth on every smaller cylinder, with local estimates inherited from [Lemmas 5.4](#) and [5.7](#).

Proof. Rearrange the original Stokes equation and subtract ∇p_{near} . Every term on the right is smooth on the smaller cylinder. This determines the gradient of the harmonic tail without constructing the pressure outside the cutoff ball. The additive function $c(t)$ is immaterial. $\square$

Proof of [Theorem 5.3](#). Fix an arbitrary compact cylinder and two buffered cylinders around it. [Lemma 5.7](#) gives smoothness of the original velocity u , not of a quotient representative. Its only possible failure branch is the nonzero finite-energy harmonic field excluded by [Lemma 5.6](#).

Apply [Lemma 5.4](#) on the buffered spatial balls. The compact-source pressure is smooth. By [Lemma 5.8](#), the harmonic tail has a smooth gradient on the target cylinder. Hence ∇p is smooth there. The target cylinder was arbitrary, which proves the theorem throughout $\mathbb{R}^3 \times (0, T)$. $\square$

6. FULL-SCALE NAVIER-STOKES INSTANTIATIONS

The Stokes graph is small enough to inspect on one page. The same method is used in three companion Navier-Stokes developments whose joint regularity consequence is stated below. Each opens with a proof-first reading guide, an acyclic diagram, a node-by-node source ledger, a monotone-hypothesis ledger, and an exhaustive branch-closure table. The counts in [Table 6](#) refer to solid numbered nodes and diagram parts in those source files.

TABLE 6. Complete large-scale proof-graph instantiations.

Development	Nodes	Diagram parts	Source
Navier-Stokes setup and compactness architecture	39	2	proof_setup.tex
Type I residual-closure argument	159	9	type_I_residual_closure.tex
Type II regularity argument	135	12	type_II_regularity.tex

The first development proves local concentration, the Type I/Type II dichotomy, and the Seregin-profile reduction for the Type I branch. The second proves that the residual Type I profile class is empty. The third excludes physical local Type II singular germs. Their joint conclusion is the following assembly theorem.

Theorem 6.1 (Joint Navier-Stokes regularity consequence). *Let (u, p) be a finite-energy suitable weak solution of the three-dimensional incompressible Navier-Stokes equations*

$$\partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0,$$

on $\mathbb{R}^3 \times (0, T)$. No point $x_ \in \mathbb{R}^3$ is a singular terminal point at time T . Since $T > 0$ is arbitrary, the combined result rules out finite-time singular points.*

Proof. Suppose that $z_* = (x_*, T)$ is singular. The local concentration theorem in the setup development produces positive Caffarelli–Kohn–Nirenberg concentration and the exhaustive alternative between a local pointwise Type I bound

$$\operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} < \infty$$

for some $\rho > 0$, and its failure, the local Type II alternative. In the Type I case, the setup development reduces z_* to a bounded centered Seregin profile with invariant local L^3 -nonvanishing. Its explicit Liouville classes are closed in the setup development, and its remaining class is empty by the residual-closure development. The Type I final assembly therefore makes z_* regular, a contradiction. In the complementary case, z_* is a physical local Type II singular germ, which is excluded by the Type II development. Both alternatives are impossible. $\square$

The node and diagram counts quantify three complete instantiations of the representation. The graphs expose profile and scale alternatives, pressure gauges, compactness topologies, recurrence routes, and terminal closure sources. At this scale, branch-preserving repair retains a large verified continuation on the positive arm of a failed intermediate assertion while adding its complementary case as a new branch.

The source repository also contains the broader branch-closure development from which the present PDE specialization is derived [DB26]. The present paper isolates the human–AI protocol, the repair theorem, and the PDE state variables needed to apply it.

7. DISCUSSION

7.1. What the method guarantees. If the mathematical evidence on every edge is correct, the graph guarantees that no admissible case has been omitted. If some leaves remain open, [Theorem 3.2](#) identifies the exact residual theorem. If an asserted property fails, [Theorem 3.3](#) preserves the conditional mathematics below that property and opens its complement. These guarantees are semantic and do not depend on a particular model or proof assistant.

The representation also supports useful automated checks: node identifiers can be compared with ledger rows; terminal shapes can be checked for outgoing edges; decision nodes can be required to list all alternatives; source links can be resolved; and state fields can be compared across transitions. Such checks detect structural omissions. They do not decide whether a cited PDE estimate is applicable.

7.2. Limitations. The method does not ensure that the graph is finite before the mathematics is understood, nor that every complementary branch is tractable. A poor state choice can generate unhelpful cases. Interface-respecting descendants are essential: if a later proof relied on an unstated detail of a failed argument, that later proof must first be rewritten against an explicit interface. The approach therefore trades some expository overhead for precise repair and review boundaries.

The Stokes example is linear and its first-failure contradiction is explicit. The Navier–Stokes instances are substantially larger, but they do not constitute a controlled user study. Measuring whether the protocol improves correctness, review time, or AI contribution quality is future work. A useful evaluation would compare ordinary prose development with branch-complete development on the same PDE problems and record omitted cases, repair cost, and expert review time.

8. CONCLUSION

Branch closure gives human and AI collaborators a common unit of work: a local analytic case with a declared interface and an exhaustive continuation. Its most important consequence is a different account of failure. A broken step does not destroy its descendants. It induces a dichotomy at the last verified state; the old proof survives on the positive arm, and the negative arm becomes

the new mathematical problem. Soundness follows from exhaustive routing, and partial progress has an exact semantic meaning.

The Stokes example shows why this matters even in a linear PDE: a local pressure split leaves a harmonic tail, while vorticity smoothing leaves a parasitic velocity mode. The finite-energy branch excludes that mode by an expanding-ball Caccioppoli contradiction, after which the equation closes the pressure tail locally. The accompanying Navier–Stokes developments show that the same discipline scales to proof graphs with more than one hundred states. This provides a concrete basis for AI assistance in PDE proof construction without conflating structured collaboration with automatic verification.

STATEMENT ON AI ASSISTANCE

Language models were used during manuscript editing and for consistency checks. The author selected the mathematical claims, verified the analytic arguments, and is responsible for the final text. No model output is treated as mathematical evidence merely by virtue of being generated.

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Email address: guillem@fragile.tech